

$$y = \left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} + \frac{x^2}{2}$$

$$\frac{dy}{dx} = p$$

$$y = p^2 - xp + \frac{x^2}{2}$$

$$\frac{dy}{dx} = 2p \frac{dp}{dx} - x \frac{dp}{dx} - p + x$$

$$p = 2p \frac{dp}{dx} - x \frac{dp}{dx} - p + x$$

$$2p - x = (2p - x) \frac{dp}{dx}$$

$$\frac{dp}{dx} = 1$$

$$dp = dx$$

$$p = x + c$$

$$y = (x+c)^2 - x(x+c) + \frac{x^2}{2}$$

$$y = \cancel{x^2} + 2xc + c^2 - \cancel{x^2} - xc + \frac{x^2}{2}$$

$$y = xc + \frac{x^2}{2} + c^2$$

$$8y.(y')^2 - 2x.y' + y = 0$$

$$8y\left(\frac{dy}{dx}\right)^2 - 2x\frac{dy}{dx} + y = 0$$

$$p = \frac{dy}{dx}$$

$$8yp^2 - 2xp + y = 0$$

$$1) \quad y - x \frac{dy}{dx} = \left(\frac{dy}{dx}\right)^2$$

$$y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 \quad p = \frac{dy}{dx}$$

$$y = xp + p^2$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + 2p \frac{dp}{dx}$$

$$\cancel{p} = x \frac{dp}{dx} + \cancel{p} + 2p \frac{dp}{dx}$$

$$0 = x \frac{dp}{dx} + 2p \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x + 2p)$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = xc + c^2$$

$$2) \quad y + \left(\frac{dy}{dx}\right)^2 = x \frac{dy}{dx} \quad p = \frac{dy}{dx}$$

$$y = x \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^2$$

$$y = xp - p^2$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p - 2p \frac{dp}{dx}$$

$$p = x \frac{dp}{dx} + p - 2p \frac{dp}{dx}$$

$$dp = 0$$

$$p = c$$

$$y = xc - c^2$$

$$3) y = x \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$p = \frac{dy}{dx}$$

$$y = xp + \sqrt{1 + p^2}$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + \frac{1}{2} (1 + p^2)^{-1/2} \cdot 2p$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + \frac{p}{\sqrt{1 + p^2}}$$

$$p = x$$

$$4) y = x \frac{dy}{dx} + \frac{dy}{dx}$$

$$\frac{dy}{dx} = p$$

$$y = xp + p$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + \frac{dp}{dx}$$

$$p = x \frac{dp}{dx} + p + \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x + 1)$$

$$dp = 0$$

$$p = c$$

$$y = xc + c$$

$$5) y = x \frac{dy}{dx} + \sin\left(\frac{dy}{dx}\right)$$

$$p = \frac{dy}{dx}$$

$$y = xp + \sin p$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + \cos p \frac{dp}{dx}$$

$$p = x \frac{dp}{dx} + p + \cos p \frac{dp}{dx}$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = xc + \sin c$$